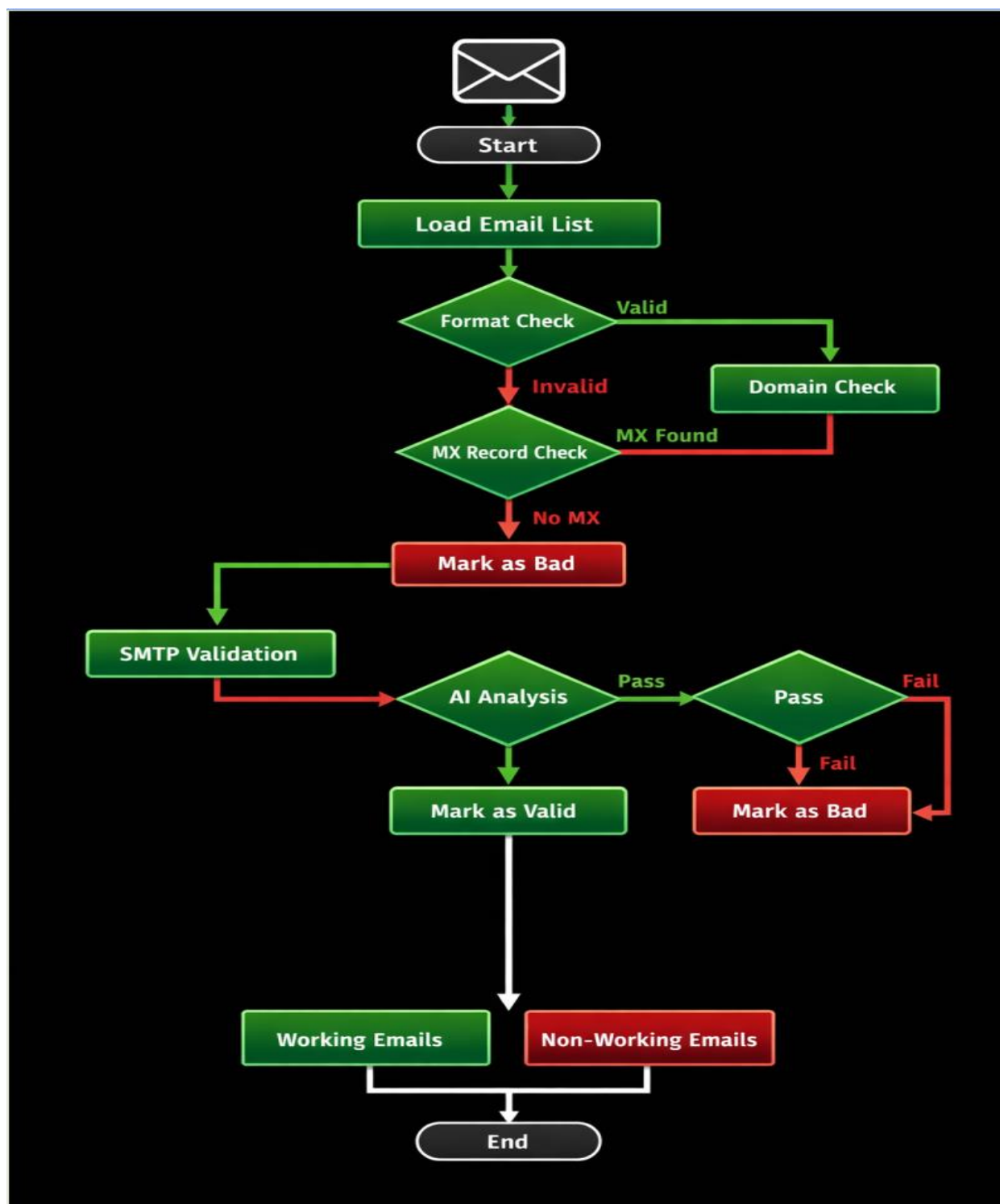
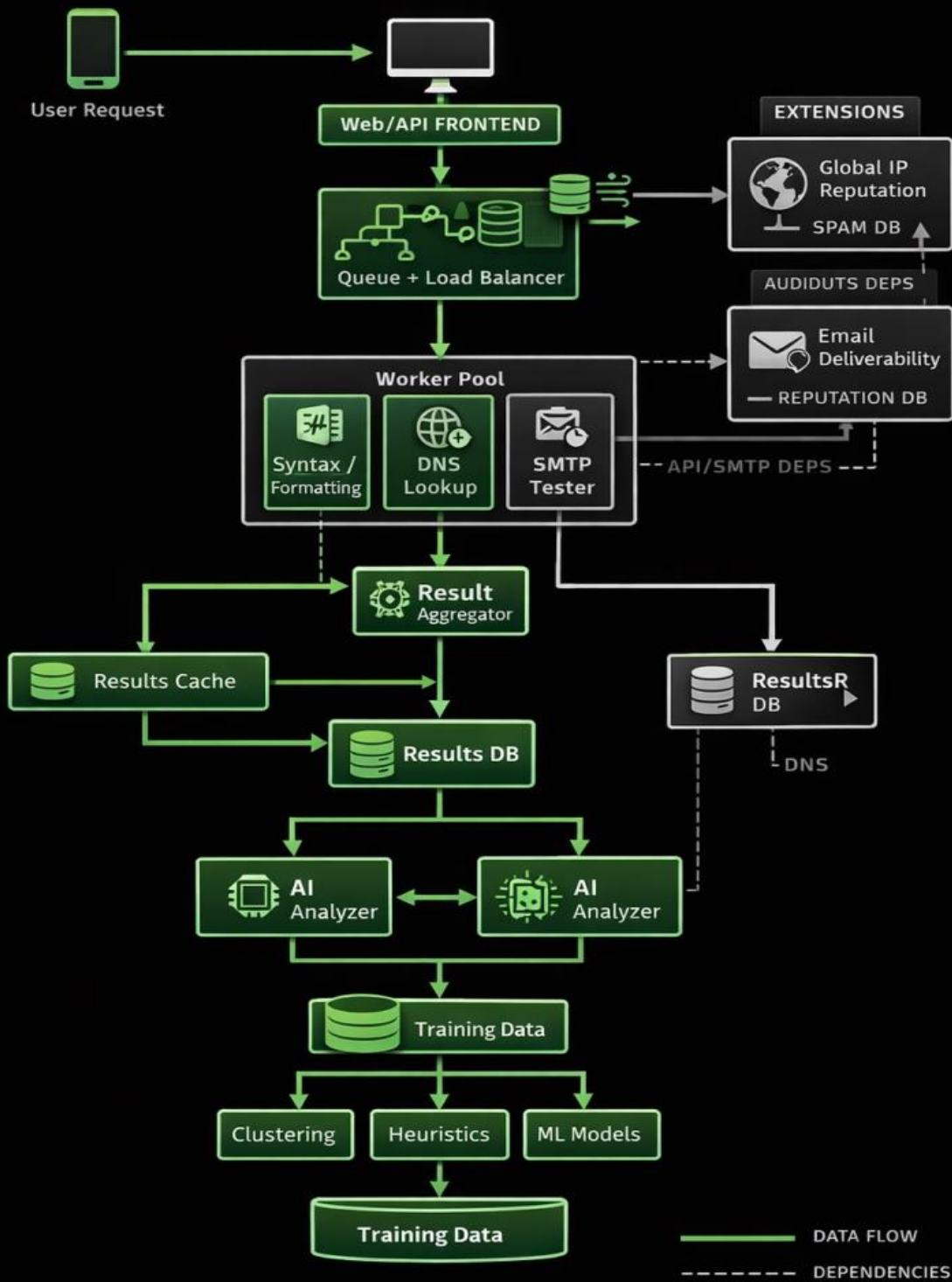


Here's a **complete, industry-grade list of parameters used in email validation logic**, from basic syntax checks to advanced SMTP & AI-based validation — **organized as a clean table**.

This matches what you've been working on already (MX lookup, SMTP 550 checks, Python + Java validators), so think of this as the **full validation checklist** used by real email-verification systems.



PRODUCTION-READY EMAIL VALIDATION ARCHITECTURE



❑ **Email Validation Parameters (Complete Table)**

#	Parameter Name	Category	What It Checks	Why It Matters
1	Email Length	Syntax	Total email length ≤ 254 chars	RFC standard compliance
2	Local Part Length	Syntax	Username ≤ 64 chars	Prevents invalid mailbox
3	Domain Length	Syntax	Domain ≤ 253 chars	DNS compatibility
4	@ Symbol Count	Syntax	Exactly one @	Core email structure
5	Allowed Characters	Syntax	Valid ASCII / UTF-8 chars	Prevents malformed emails
6	Dot Placement	Syntax	No leading, trailing, or double dots	RFC compliance
7	Quoted Strings	Syntax	Proper quotes usage	Advanced RFC support
8	Unicode / IDN	Syntax	Non-ASCII domain support	International emails
9	Disposable Email	Policy	Temp email providers	Avoid fake signups
10	Role-Based Address	Policy	admin@, info@, support@	Low engagement / shared inbox
11	Blacklisted Keywords	Policy	spam keywords	Reduce spam risk
12	Whitelisted Domains	Policy	trusted domains	Skip heavy validation
13	Domain Format	Domain	Valid TLD & domain name	DNS correctness
14	Domain Existence	Domain	Domain resolves	Avoid fake domains
15	MX Record Presence	DNS	Mail server exists	Required for email delivery
16	MX Record Priority	DNS	Valid MX priority values	Mail routing correctness
17	MX Server Reachable	Network	Server responds	Detect dead mail servers

18	SPF Record	DNS	Sender Policy Framework	Anti-spoofing
19	DKIM Record	DNS	DomainKeys Identified Mail	Email authenticity
20	DMARC Record	DNS	Domain Message Auth	Domain policy enforcement
21	A / AAAA Record	DNS	IP resolution	Fallback routing
22	SMTP Connection	SMTP	Port 25 / 587 reachable	Mail server availability
23	SMTP HELO/EHLO	SMTP	Server greeting	SMTP protocol compliance
24	SMTP MAIL FROM	SMTP	Sender accepted	Server trust check
25	SMTP RCPT TO	SMTP	Recipient accepted	Mailbox existence
26	SMTP Response Code	SMTP	250 / 450 / 550 etc.	Detect valid/invalid mailbox
27	Catch-All Domain	SMTP	Accepts all recipients	Ambiguous validity
28	Greylisting	SMTP	Temporary rejection	Retry logic needed
29	Mailbox Full	SMTP	Storage exceeded	Delivery failure risk
30	Server Throttling	SMTP	Rate limits	Avoid IP blocking
31	SMTP Timeout	SMTP	Response delay	Network reliability
32	SMTP TLS Support	Security	STARTTLS available	Secure transmission
33	Reverse DNS (PTR)	Security	IP ↔ domain mapping	Spam detection
34	IP Reputation	Security	Blacklisted IPs	Deliverability risk
35	Domain Reputation	Security	Historical spam behavior	Trust score
36	Email Age	Heuristic	Domain/email lifespan	Fraud detection
37	Historical Bounce Rate	Heuristic	Previous failures	Predict reliability
38	Engagement Signals	Heuristic	Opens / clicks	Human usage
39	Typo Detection	Heuristic	gmial → gmail	Auto-correction

40	MX Provider Type	Heuristic	Gmail, Outlook, Custom	Behavior prediction
41	Country of Domain	Heuristic	Geo-location	Compliance & risk
42	Frequency Pattern	Heuristic	Mass-generated emails	Bot detection
43	AI Risk Score	AI	ML-based validity	Advanced filtering
44	AI Pattern Matching	AI	Known fake patterns	Fraud prevention
45	Behavioral Similarity	AI	Similar email clusters	Bot networks
46	Email Hash Uniqueness	AI	Duplicate detection	Fake registrations
47	Validation Confidence	Output	Final confidence score	Decision making
48	Validation Status	Output	Valid / Invalid / Risky	Business logic
49	Failure Reason	Output	Exact rejection cause	Debugging & logs
50	Validation Timestamp	Output	Time of check	Auditing
51	Validation Source	Output	Syntax/DNS/SMTP/AI	Transparency
52	Retry Eligibility	Output	Can retry later?	Queue management
53	Data Privacy Flag	Compliance	GDPR safe handling	Legal compliance
54	Logging Level	System	What to log	Security & debugging
55	Thread / Rate Control	System	Parallel validation	Performance
56	Proxy / IP Rotation	System	Avoid SMTP bans	Scalability
57	Result Persistence	System	DB/File storage	Reporting
58	Validation Version	System	Logic version	Backward compatibility

□ Practical Usage Levels

Use Case	Required Parameters
Simple Form Validation	1-8
Signup Protection	1-15

Bulk Email Cleaning	1-30
Enterprise Validation	1-42
AI-Based Anti-Fraud	1-46
Production SaaS Tool	All

❑ **Pro Tip (From Real-World Systems)**

✓ **Never rely only on SMTP 550**

✓ **Combine Syntax + DNS + SMTP + Heuristics**

✓ **Add AI scoring** instead of hard “valid/invalid”

✓ **Treat catch-all domains** as “Risky”, not valid

If you want next:

- ✓ **Flow diagram of email validation**
- ✓ **Java vs Python implementation mapping**
- ✓ **Production-ready validation architecture**
- ✓ **Sample DB schema for storing validation results**

Just tell me ❑